

Avery Wong

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EDUCATION

University of Toronto Toronto, ON
Bachelor of Applied Science in Mechanical Engineering + PEY Co-op Sept 2024 - May 2029 (expected)
Minor in Engineering Business
Relevant Coursework: Engineering Strategies & Practice, Statistics, Mechanical Engineering Design, Manufacturing, Numerical Methods, Computer Programming, Thermodynamics

SKILLS

Engineering & Programming Tools: SolidWorks, ANSYS, MATLAB, Python, Minitab, HTML/CSS
Productivity Tools: Google Workplace (Sheets, Slides), Microsoft Office (Excel, PowerBI, Project, Powerpoint)
Professional Skills: Project Management, Communication, Collaboration, Troubleshooting, Adaptability

EXPERIENCE

Mechanical and User Experience Lead Toronto, ON
University of Toronto Biomedical Design Team Aug 2025 – May 2026

- Led research and development (R&D) efforts to create a smart rehabilitation knee sleeve that tracks muscle activation and range of motion in users that are recovering from ACL tears.
- Applied CES EduPack software and market benchmarking to identify and filter candidate materials based on multiple performance criteria, shortlisting materials for further prototype testing and analysis.
- Designed and modeled geometric prototypes using SolidWorks and conducted physical tests with sample materials and stress analysis simulations through ANSYS to determine final material composition, resulting in a triple-layered material mesh that balances cost and performance.
- Troubleshot poor surface electromyography (sEMG) signals by increasing knee sleeve compression, optimizing power input, reducing electromagnetic interference, and introducing a grounding electrode, resulting in a 50% improvement in signal quality.

Engineering Lead Toronto, ON
Mechanical Engineering Design - University of Toronto Sept 2025 – Dec 2025

- Spearheaded the design and development of a low-cost, long reach 6-DOF robotic arm for cinematography applications.
- Facilitated weekly team meetings to assign tasks and set project timelines using Trello and Google Workspace, ensuring on-schedule task delivery across the team.
- Designed and modeled a fully constrained digital prototype using SolidWorks, guided by a 1 m/s speed and 5kg payload requirement, validating technical feasibility.
- Converted the digital prototype into detailed SolidWorks assembly drawings and Bills of Materials (BOM) across multiple subassemblies, specifying critical dimensions for non-standardized components to facilitate future machining.

Team Lead Toronto, ON
Engineering Strategies & Practice II - University of Toronto Sept 2024 – May 2025

- Led a team of 6 to conceptually design and develop an electrical outlet extender for use in an elementary school classroom, successfully delivering a prototype within strict time constraints.
- Conducted market research and applied ISO Standards and ESA Codes to generate over 75 unique design solutions, ensuring full compliance with client and stakeholder requirements.
- Facilitated weekly team meetings to assign tasks through a Gantt Chart and update the project timeline using Microsoft Project, ensuring all milestones were delivered on schedule.
- Co-authored an IEEE-formatted technical report and PowerPoint presentation, successfully demonstrating how the proposed design met all the required specifications.

ADDITIONAL

Certifications & Training: Certified SolidWorks Associate (CWSA), Basic Machining at George Brown College
Websites: <https://avery-06.github.io/portfolio/>